

Electronic Circuits Quiz Answer Key

1. What is the primary function of Raspberry Pi?

- ☐ To control servo motors
- ☒ **To run Python programs**
- ☐ To play video games
- ☐ To connect to the internet

2. What size SD card is recommended for installing Raspberry Pi OS?

- ☐ 4GB
- ☒ **8GB**
- ☐ 16GB
- ☐ 32GB

3. What is the purpose of PWM pins on Arduino?

- ☒ **To read analog sensors**
- ☐ To control servo motors
- ☐ To display graphics
- ☐ To connect to the internet

4. Which Raspberry Pi model offers onboard WiFi and Bluetooth connectivity?

- ☐ Raspberry Pi Zero
- ☒ **Raspberry Pi 3 Model B+**
- ☐ Raspberry Pi 4 Model B
- ☐ Raspberry Pi Compute Module

5. Which platform is more suitable for projects requiring low power consumption?

- ☐ Raspberry Pi
- ☒ **Arduino**
- ☐ Both consume similar power
- ☐ It depends on the specific project requirements

6. Which platform is commonly used for data logging applications?

- ☒ **Raspberry Pi**
- ☐ Arduino
- ☐ Both are equally common
- ☐ Neither

7. Which platform is commonly used for projects requiring real-time processing?

- ☐ Raspberry Pi

- ☒ **Arduino**
- ☐ Both are equally suitable
- ☐ Neither

8. Which command is used to update the software packages on Raspberry Pi?

- ☐ sudo apt update
- ☐ sudo apt upgrade
- ☐ sudo apt install
- ☒ **sudo apt update && sudo apt upgrade**

9. What does the acronym "IDE" stand for in Arduino IDE?

- ☐ Interactive Design Environment
- ☐ Intelligent Development Environment
- ☒ **Integrated Development Environment**
- ☐ Interface Design Environment

10. Which platform is commonly used for educational purposes in teaching programming and electronics?

- ☐ Raspberry Pi
- ☐ Arduino
- ☒ **Both are equally common**
- ☐ Neither

11. Which platform is commonly used for home automation projects?

- ☒ **Raspberry Pi**
- ☐ Arduino
- ☐ Both are equally common
- ☐ Neither

12. Which platform is commonly used for projects requiring extensive community support and resources?

- ☐ Raspberry Pi
- ☐ Arduino
- ☒ **Both are equally common**
- ☐ Neither

13. Which interface is commonly used to connect a Raspberry Pi camera module?

- ☐ USB
- ☐ HDMI
- ☒ **CSI (Camera Serial Interface)**
- ☐ I2C (Inter-Integrated Circuit)

14. What is the maximum number of digital pins available on the Arduino Uno?

- ☐ 8
- ☐ 12
- ☒ **14**
- ☐ 20

15. Which platform is more suitable for projects requiring real-time operating system (RTOS) capabilities?

- ☐ Raspberry Pi
- ☒ **Arduino**
- ☐ Both are equally suitable
- ☐ Neither

16. Which platform is more suitable for projects requiring extensive graphical user interfaces (GUIs)?

- ☒ **Raspberry Pi**
- ☐ Arduino
- ☐ Both are equally suitable
- ☐ Neither

17. Which platform is commonly used for projects requiring mobile applications integration?

- ☒ **Raspberry Pi**
- ☐ Arduino
- ☐ Both are equally common
- ☐ Neither

18. Which platform is commonly used for projects requiring machine learning and artificial intelligence capabilities?

- ☒ **Raspberry Pi**
- ☐ Arduino
- ☐ Both are equally common
- ☐ Neither

19. What does this python code do?

```

1  import RPi.GPIO as GPIO
2  import time
3
4  GPIO.setmode(GPIO.BCM)
5  led_pin = 18
6  GPIO.setup(led_pin, GPIO.OUT)
7
8  try:
9      while True:
10         GPIO.output(led_pin, GPIO.HIGH)
11         time.sleep(1)
12         GPIO.output(led_pin, GPIO.LOW)
13         time.sleep(1)
14 except KeyboardInterrupt:
15     GPIO.cleanup()

```

- ☐ Reads data from a temperature sensor and displays it on an LED.
- ☐ Controls a servo motor's position based on sensor input.
- ☒ **Blinks a LED at a 1-second interval.**
- ☐ Will display syntax errors
- ☐ None of the above.

20. What does this python code do?

```

1  import RPi.GPIO as GPIO
2  import time
3
4  pin1 = 17
5  pin2 = 18
6
7  GPIO.setmode(GPIO.BCM)
8  GPIO.setup(pin1, GPIO.IN, pull_up_down=GPIO.PUD_DOWN)
9  GPIO.setup(pin2, GPIO.OUT)
10
11 try:
12     while True:
13         if GPIO.input(pin1) == GPIO.HIGH:
14             GPIO.output(pin2, GPIO.HIGH)
15         else:
16             GPIO.output(pin2, GPIO.LOW)
17 except KeyboardInterrupt:
18     GPIO.cleanup()

```

- ☐ Reads data from a temperature sensor and toggles an LED accordingly.
- ☒ **Blinks an LED when a button is pressed.**
- ☐ Controls a servo motor's position based on button input.
- ☐ Will display syntax errors
- ☐ None of the above.

21. What is the purpose of the KeyboardInterrupt exception in Python?

- ☐ It is raised when there is an error in keyboard input.
- ☐ It is raised when a program encounters an unexpected error.
- ☒ **It is raised when a user interrupts the execution of a program by pressing CTRL+C.**
- ☐ It is raised when there is an error in handling keyboard events.

22. What does this python code do?

```

1  from gpiozero import RGBLED
2  from time import sleep
3  pin1 = 17
4  pin2 = 28
5  pin3 = 22
6  led = RGBLED(red=pin1, green=pin2, blue=pin3)
7  try:
8      led.color = (1, 0, 0)
9      sleep(1)
10     led.color = (0, 1, 0)
11     sleep(1)
12     led.color = (0, 0, 1)
13     sleep(1)
14     led.off()
15 except KeyboardInterrupt:
16     led.off()

```

- ☐ Blinks an RGB LED at a 1-second interval.
- ☐ Alternately lights up the three colors of an RGB LED at a 1-second interval.
- ☐ Lights up red and blue at a 1-second interval.
- ☒ **Display error message**
- ☐ None of the above

23. What does this C code do?

```
1  int basePin = 9;
2  int v;
3  void setup()
4  ▼ {
5      pinMode(basePin, OUTPUT);
6      Serial.begin(9600);
7  }
8  void loop()
9  ▼ {
10     if(Serial.available())
11     ▼ {
12         v = Serial.parseInt();
13         analogWrite(basePin, v);
14     }
15 }
```

- ☐ Toggles Light
- ☐ Control the DC motor speed
- ☒ **May not work because of values errors**
- ☐ May not work because of syntax errors

24. What does this C code do?

```

1  int greenPin = 5;
2  int bluePin = 6;
3  int redPin = 3;
4  void setup()
5  ▼ {
6      pinMode(greenPin, OUTPUT);
7      pinMode(bluePin, OUTPUT);|
8      pinMode(redPin, OUTPUT);
9  }
10 void loop()
11 ▼ {
12     SetColor(255,0,0);
13     delay(1000);
14     SetColor(0,255,0);
15     delay(1000);
16     SetColor(0,0,255);
17     delay(1000);
18     SetColor(255,255,255);
19     delay(1000);
20     SetColor(170,0,255);
21     delay(1000);
22     SetColor(50,100,90);
23     delay(1000);
24 }

```

- ☐ Bliks RGB LED
- ☐ Toggles Light
- ☐ May not work because of values errors
- ☒ **May not work because of syntax errors**